

PocketSmart AI

Full VS Code Development Specification & AI Master Prompt Document

□ **Document Purpose:** Provides a complete, end-to-end master prompt and technical architecture guide to build the PocketSmart AI web application from scratch in Visual Studio Code, leveraging Gemini 1.5 Flash / Pro and multi-platform e-commerce integrations.

1. Executive Summary & Concept Overview

Managing cross-category budgets across various life events—such as home renovation, wedding or party planning, fashion styling, and travel—is often fragmented and stressful. **PocketSmart AI** bridges this gap using Generative AI (Gemini 1.5 Flash / Pro) to curate personalized, budget-aligned recommendations across top e-commerce and service platforms including Amazon, IKEA, Flipkart, Swiggy, Zomato, OYO, Myntra, and Pepperfry.

2. Master Prompt for Cursor / VS Code Copilot

Copy and paste the exact prompt below into VS Code (Cursor AI, GitHub Copilot Workspace, or ChatGPT) to build the complete full-stack project structure:

```
You are an expert Principal Full-Stack AI Engineer. Create a complete, production-ready web application named 'PocketSmart AI' from scratch.

### CONCEPT OVERVIEW:
PocketSmart AI is a GenAI-powered, cross-platform budget recommendation engine. It helps users manage budgets across diverse life needs (Home Decor, Event/Party Planning, Jewelry & Fashion Matching, Custom Goal/Trip Planning) by generating personalized product and service recommendations from Amazon, Flipkart, IKEA, Swiggy, Zomato, OYO, Myntra, and Pepperfry.

### CORE FEATURES TO BUILD:
1. Multi-Category Smart Planners:
  - Home Decor: Room size, aesthetic preference (Minimalist, Bohemian, Modern, Industrial), budget limit, priority furniture.
  - Event & Party Planner: Guest count, event type (Birthday, Wedding, Dinner), food vs. venue vs. decor ratio.
  - Jewelry & Fashion Matcher: Outfit color/style, event type, metal preference (Gold, Silver, Artificial), price cap.
  - Custom Goal / Trip Planner: Flexible multi-category input.
```

- 2. Real-Time Budget Visualizer:
 - Dynamic Chart.js allocation breakdown (Category vs. Expense).
 - Interactive Budget Gauge / Meter (Safe, Caution, Over-Budget warnings).
- 3. Gemini AI Recommendation Engine (Gemini 1.5 Flash / Pro Integration):
 - Direct API connectivity with client-side key config.
 - Intelligent prompt grounding with realistic cross-platform product items, pricing, and buy links.
 - Offline Simulation Mode fallback with rich mock data when API key is not provided.
- 4. Cross-Platform Ecosystem Hub:
 - Unified product cards with source badges (Amazon, IKEA, Swiggy, etc.), real-time pricing, ratings, and direct search/buy links.
- 5. Interactive Wishlist & Shopping Plan Manager:
 - Add items to plan, total budget tracking, over-budget alert banners, item priority tagging, and JSON / Print export.
- 6. Design System:
 - Modern Glassmorphism Fintech SaaS design using Tailwind CSS, dark/light theme toggle, custom UI components, responsive sidebar navigation.

3. Technical Architecture & Tech Stack

Layer	Technology / Library	Role & Function
Frontend Framework	HTML5, Tailwind CSS, JavaScript (ES6+)	UI components, responsive layouts, glassmorphism design
AI / LLM Core	Google Gemini 1.5 Flash / Pro API	Contextual parsing, recommendation generation, budget optimization
Data Visualization	Chart.js	Visualizing category breakdown charts and budget gauge gauges
Icons & UI Graphics	FontAwesome 6.4 / Lucide Icons	Platform icons, action buttons, category visual cues
State Management	Browser LocalStorage & JS State	Wishlist caching, custom budget presets, active API key persistence

4. Project Folder Structure in VS Code

```
pocketsmart-ai/
├── index.html           # Main Single-Page Application (SPA)
├── css/
│   └── custom.css      # Custom glassmorphism styles and animations
├── js/
│   ├── app.js          # Application state, UI routing, local storage
│   ├── gemini.js       # Gemini 1.5 API client & prompt builder
│   ├── charts.js       # Chart.js budget allocation renderers
│   └── mockData.js     # Fallback multi-platform product catalog
└── assets/
    └── images/         # Platform logos and hero banners
```

5. Step-by-Step VS Code Setup Guide

Step 1: Environment Setup

1. Open VS Code and create a new directory named **pocketsmart-ai**.
2. Install the **Live Server** extension in VS Code for live browser reloads.
3. Create an **index.html** file at the root.

Step 2: Key API Configuration

PocketSmart AI connects directly to Google AI Studio using Gemini 1.5 Flash or Pro. Obtain an API key from **aistudio.google.com** and enter it into the application's top API settings drawer. If no key is supplied, the app seamlessly activates its offline recommendation simulator.

Step 3: Execution & Testing

Right-click **index.html** in VS Code and click **Open with Live Server**. Test across the four smart planners and add recommended items from Amazon, IKEA, and Swiggy into the live budgeting cart.